KurveRSC: select the relational program, then its fitting procedure

Schema-derived graphs · temporal validation · pluggable downstream learner (CatBoost here)

1 Data + prediction task

Root: customer · predict 30-day churn

customer

id	region
C1	US
C2	EU

order

oid	cid	time	total
O1	C1	t−7d	80
O2	C2	t−2d	50

product

pid	type
P1	A
P2	B

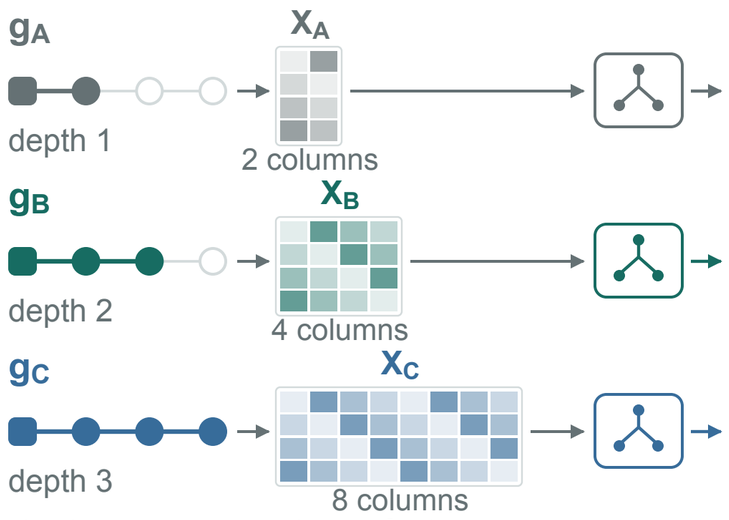
order_line

oid	pid	value
O1	P1	30
O2	P2	50



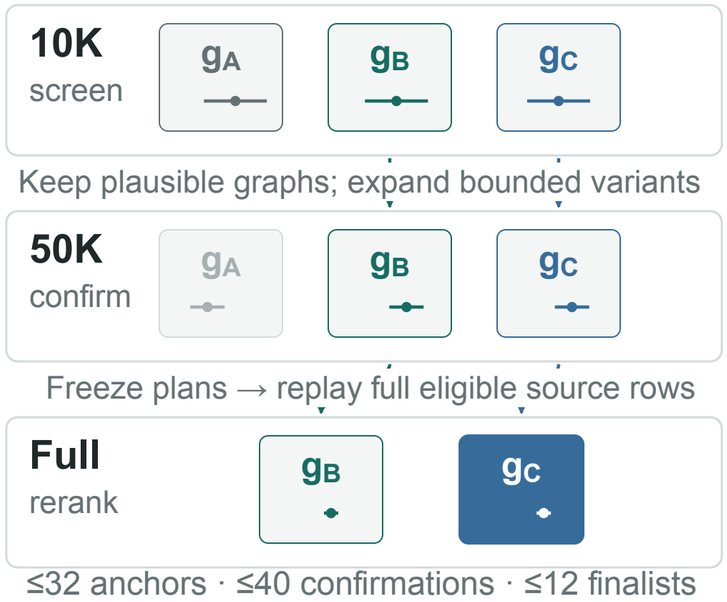
2 Programs → feature frames

GraphReduce: filter → annotate → reduce → join



Root payload + relational features; same learner.

3 S1 · Promote plausible graphs



All comparisons are temporal: sources precede each query; forecast labels remain masked.

4 S2 · Check production fitting

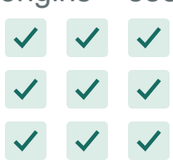


Same past-to-future test for both graphs

Earlier labels + **Newly known** → **Held-out block**

initial fit | refit support | predict + score

origins × seeds



All seeds choose smaller

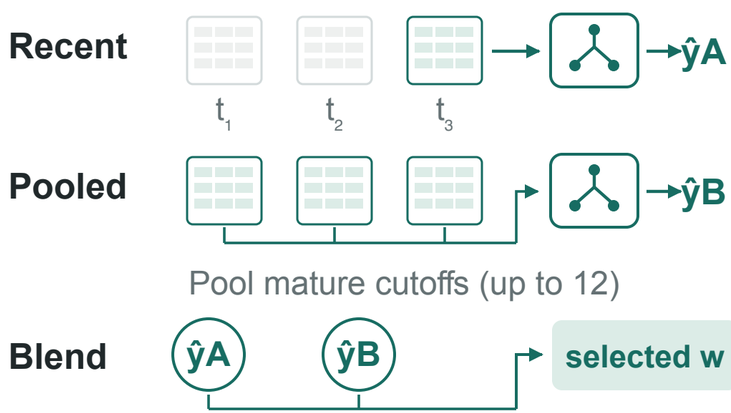
Every compared loss ≤1.01× the incumbent loss

L = 1−AUC or MAE

One graph g*: pass → g_B; else → g_C

5 S3 · Select training support

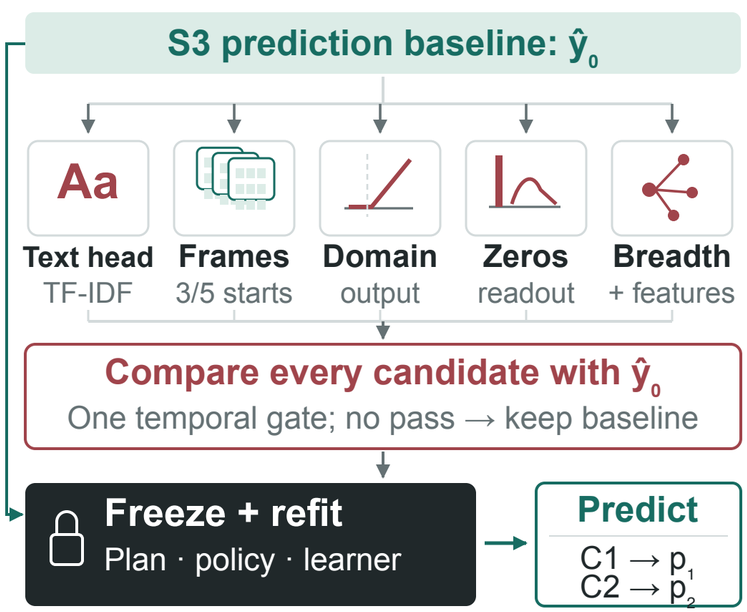
Keep g* fixed; vary the fitting policy



Temporal validation chooses the fitting recipe.

g* + selected policy → S3 control $\hat{y}_0$

6 S4 · Challenge, then freeze



Schematic: feature widths and score intervals illustrate decisions; budgets and stage roles follow KurveRSC 0.2.0.